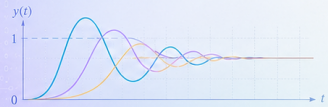
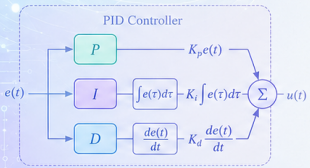
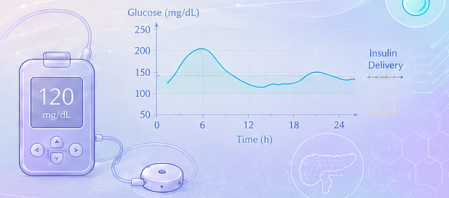
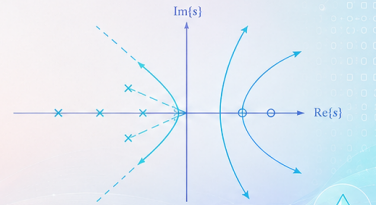
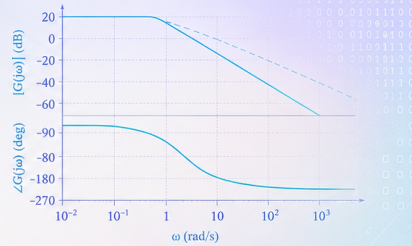


Time Response

SAHAJ SAXENA
sahajsaxena.github.io



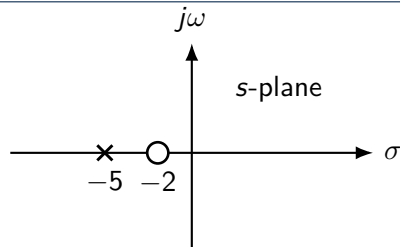
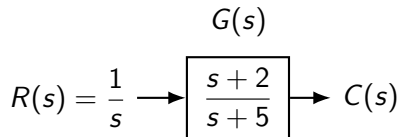
$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



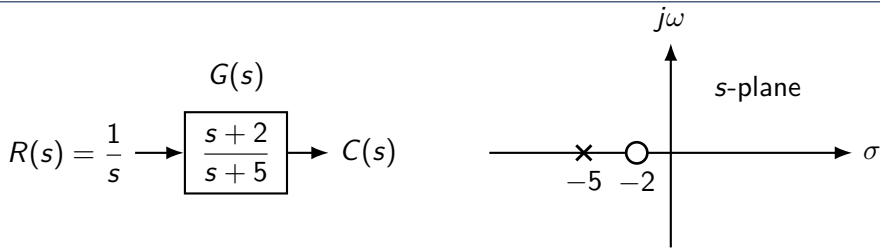
LTI System Response and Convolution Equation

- ▶ The output response of a system is the **sum of two responses**: the forced response and the natural response
- ▶ The forced response is also called the **steady-state response** or particular solution. The natural response is also called the homogeneous solution.
- ▶ The **poles** of a transfer function are the values of the Laplace transform variable, s , that cause the transfer function to become **infinite**
- ▶ The **zeros** of a transfer function are the values of the Laplace transform variable, s , that cause the transfer function to become **zero**

Poles and Zeros of a First-Order System: An Example



Poles and Zeros of a First-Order System: An Example

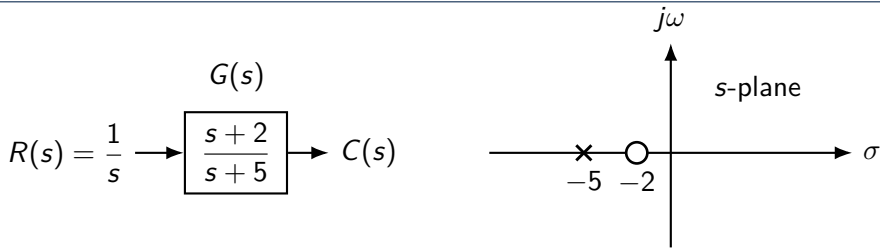


$$C(s) = R(s)G(s) = \frac{s+2}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5}.$$

where

$$A = \left. \frac{s+2}{s+5} \right|_{s=0} = \frac{2}{5}, \quad B = \left. \frac{s+2}{s} \right|_{s=-5} = \frac{3}{5}.$$

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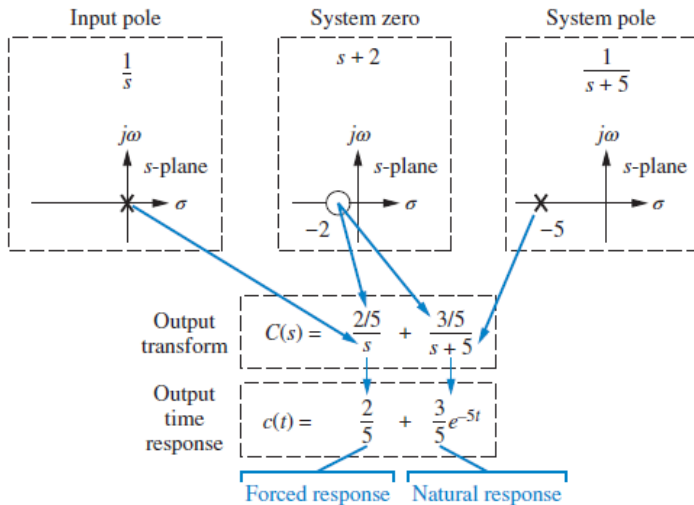
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Thus,

$$c(t) = \frac{2}{5} + \frac{3}{5}e^{-5t}$$

Observations:



Interpretation of Poles and Zeros

- ▶ A **pole of the input** determines the form of the **forced response**.
For example, a pole at the origin due to a step input generates a constant term in the output response.
- ▶ A **pole of the transfer function** determines the form of the **natural response**.
For example, the pole at

$$s = -5$$

generates the exponential term

$$e^{-5t}.$$

- A real pole at

$$s = -a$$

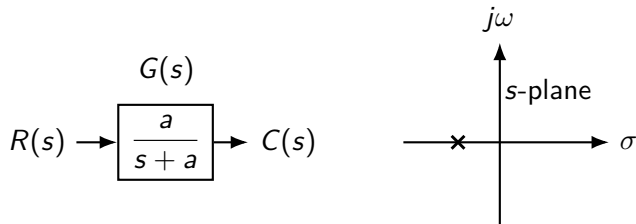
produces a transient response of the form

$$e^{-at}.$$

Therefore, the farther a stable pole lies to the left in the s -plane, the **faster the transient response decays**.

- The locations of the **poles and zeros** also determine the amplitudes of the forced and natural response components through the coefficients obtained in the partial-fraction expansion.

First-Order System Without Zeros



- Consider a **first-order system without zeros**:

$$G(s) = \frac{a}{s+a}, \quad a > 0.$$

- It has a single pole at

$$s = -a.$$

- This model is commonly used to describe the dynamics of thermal, electrical, and mechanical systems with a single dominant energy-storage mode.

Step Response of a First-Order System

- For a unit-step input,

$$R(s) = \frac{1}{s}.$$

- The Laplace transform of the output is

$$C(s) = R(s)G(s) = \frac{1}{s} \cdot \frac{a}{s+a} = \frac{a}{s(s+a)}.$$

- Using partial-fraction expansion,

$$\frac{a}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a}.$$

- Solving for the constants:

$$a = A(s+a) + Bs.$$

Hence,

$$A = 1, \quad B = -1.$$

Time-Domain Step Response

Therefore,

$$C(s) = \frac{1}{s} - \frac{1}{s+a}.$$

Taking the inverse Laplace transform,

$$c(t) = \mathcal{L}^{-1}\{C(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s+a}\right\}.$$

Hence,

$$c(t) = 1 - e^{-at}, \quad t \geq 0$$

- ▶ The response can be written as

$$c(t) = c_f(t) + c_n(t),$$

where

$$c_f(t) = 1 \quad \text{and} \quad c_n(t) = -e^{-at}.$$

- ▶ The term $c_f(t)$ is the **forced response** generated by the step input.
- ▶ The term $c_n(t)$ is the **natural response** generated by the system pole at $s = -a$.
- ▶ As $t \rightarrow \infty$,

$$e^{-at} \rightarrow 0 \quad \Rightarrow \quad c(t) \rightarrow 1.$$

Interpretation of Pole Location

- ▶ A real pole at

$$s = -a$$

generates an exponential term of the form

$$e^{-at}.$$

- ▶ Therefore, the farther the pole is to the left in the s -plane, the larger the value of a , and the **faster** the transient response decays.
- ▶ For the first-order system,

$$c(t) = 1 - e^{-at}.$$

- ▶ The only parameter governing the transient response is a .
- ▶ A convenient performance measure is the **time constant**

$$\tau = \frac{1}{a}.$$

Time Constant and Performance Specification

- At

$$t = \tau = \frac{1}{a},$$

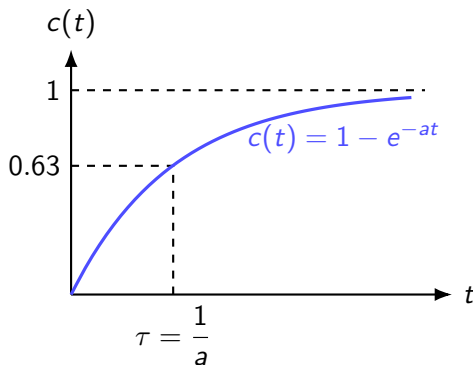
the natural response becomes

$$e^{-at} \Big|_{t=1/a} = e^{-1} \approx 0.37.$$

- Hence,

$$c(t) \Big|_{t=1/a} = 1 - e^{-1} \approx 0.63.$$

- Thus, after one time constant, the output reaches about 63% of its final value.



Rise Time, T_r for the first-order system, $c(t) = 1 - e^{-at}$.

- The **rise time** T_r is the time required for the response to rise from 10% to 90% of its final value.

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- ▶ At $c(t) = 0.1$:

$$0.1 = 1 - e^{-at}$$

$$t_{10} = -\frac{\ln(0.9)}{a} \approx \frac{0.105}{a}.$$

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- At $c(t) = 0.9$:

$$0.9 = 1 - e^{-at}$$

$$t_{90} = -\frac{\ln(0.1)}{a} \approx \frac{2.303}{a}.$$

- Therefore,

$$\begin{aligned} T_r &= t_{90} - t_{10} \\ &\approx \frac{2.303 - 0.105}{a} \approx \frac{2.2}{a}. \end{aligned}$$

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Rise Time

$$\boxed{T_r \approx \frac{2.2}{a} = 2.2\tau}, \quad \tau = \frac{1}{a}.$$

Settling Time, T_s for the first-order system, $c(t) = 1 - e^{-at}$.

- ▶ The **settling time** T_s is the time required for the response to reach and remain within a specified tolerance of its final value.

Settling Time, T_s for the first-order system, $c(t) = 1 - e^{-at}$.

- ▶ The **settling time** T_s is the time required for the response to reach and remain within a specified tolerance of its final value.

- ▶ For the 2% criterion,
$$c(T_s) = 0.98.$$

Hence,

$$0.98 = 1 - e^{-aT_s},$$

which gives

$$e^{-aT_s} = 0.02.$$

Taking the natural logarithm,
$$-aT_s = \ln(0.02).$$

Therefore,

$$\begin{aligned} T_s &= -\frac{\ln(0.02)}{a} \\ &= \frac{3.912}{a} \approx \frac{4}{a}. \end{aligned}$$

Since

$$\tau = \frac{1}{a},$$

$$\boxed{T_s \approx 4\tau}.$$

Interpretation

A larger a moves the pole farther to the left in the s -plane, resulting in a **faster response** and a smaller settling time.

Second-order System

- ▶ Whereas varying a first-order system's parameter simply **changes the speed of the response**, changes in the parameters of a second-order system can **change the form of the response**.
- ▶ For example, a second-order system can display characteristics much like a first-order system, or, depending on component values, display damped or pure oscillations for its transient response.

Parameters of second-order system

Consider a generalized second-order system of the form

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Natural Frequency, ω_n

- ▶ The *natural frequency* of a second-order system is the **frequency of oscillation** of the system without damping.
- ▶ For example, the frequency of oscillation of a series *RLC* circuit with the resistance shorted would be the natural frequency.

Damping Ratio, ζ

- ▶ a second-order system's underdamped step response is characterized by **damped oscillations**.

Standard Second-Order System

A general second-order LTI system without zeros is represented as

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

- ▶ ω_n : natural frequency
- ▶ ζ : damping ratio
- ▶ The exponential decay rate is

$$\sigma = \zeta\omega_n.$$

- ▶ Hence,

$$\zeta = \frac{\sigma}{\omega_n}.$$

The characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

Therefore, the poles are

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.$$

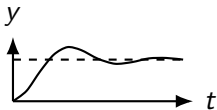
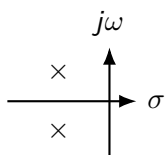
For $0 < \zeta < 1$,

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}.$$

Second-Order Poles and System Response

Underdamped: $0 < \zeta < 1$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$$

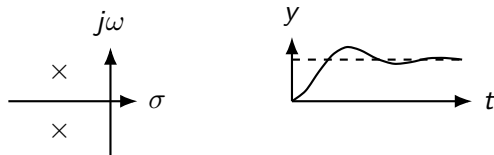


Decaying oscillatory response.

Second-Order Poles and System Response

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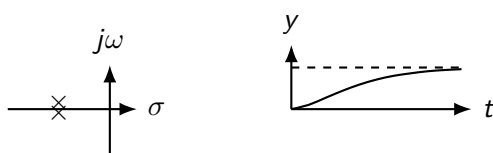
$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$



Decaying oscillatory response.

Critically damped: $\zeta = 1$

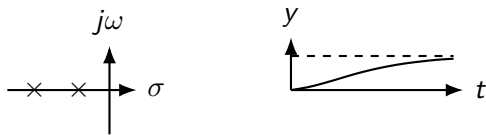
$$s_1 = s_2 = -\omega_n$$



Fastest non-oscillatory response.

Overdamped: $\zeta > 1$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$$

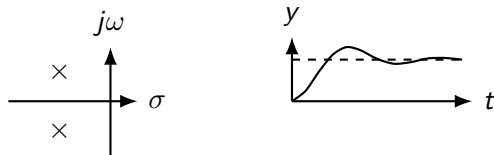


Slow, non-oscillatory response.

Second-Order Poles and System Response

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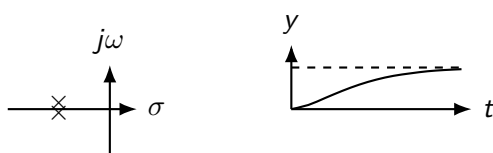
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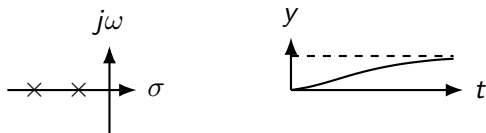
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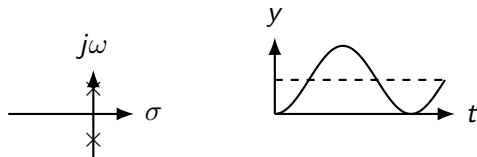
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Slow, non-oscillatory response.

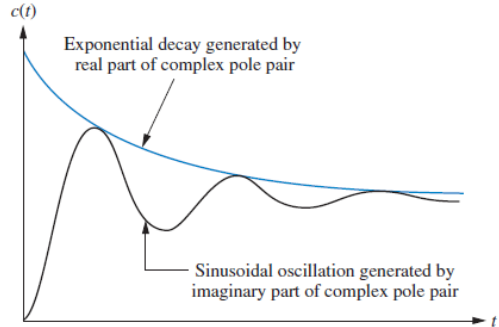
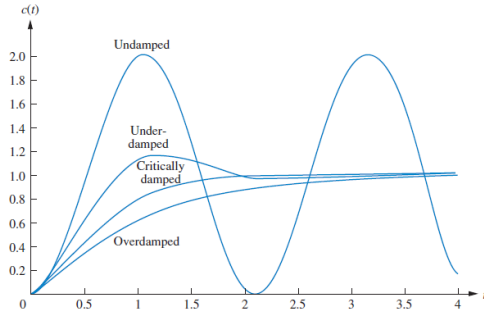
Undamped: $\zeta = 0$

$$s_{1,2} = \pm j\omega_n$$



Sustained oscillatory response.

Step responses for second-order system damping cases



Underdamped Second-Order System

For an underdamped system,

$$0 < \zeta < 1.$$

Define

$$\beta = \sqrt{1 - \zeta^2}.$$

The poles become

$$s_{1,2} = -\zeta\omega_n \pm j\beta\omega_n$$

The real part

$$-\zeta\omega_n$$

The imaginary part

$$\beta\omega_n = \omega_n\sqrt{1 - \zeta^2}$$

determines the **exponential decay rate**. determines the **damped oscillation frequency**.
Therefore,

$$\omega_d = \beta\omega_n = \omega_n\sqrt{1 - \zeta^2}.$$

Unit-Step Response of a Second-Order System

For a unit-step input, $U(s) = \frac{1}{s}$.

Therefore,

$$Y(s) = G(s)U(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}.$$

For $0 < \zeta < 1$, the inverse Laplace transform gives

$$y(t) = 1 - e^{-\zeta\omega_n t} \left[\cos(\beta\omega_n t) + \frac{\zeta}{\beta} \sin(\beta\omega_n t) \right]$$

where

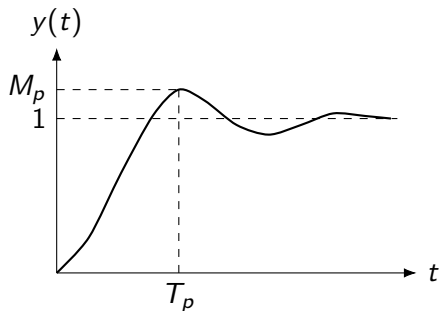
$$\beta = \sqrt{1 - \zeta^2}.$$

- ▶ $e^{-\zeta\omega_n t}$ determines the **decay envelope**.
- ▶ $\beta\omega_n$ determines the **oscillation frequency**.

Second-Order Step Response

The underdamped step response is characterized by

- ▶ Rise time, T_r
- ▶ Peak time, T_p
- ▶ Maximum overshoot, M_p
- ▶ Percent overshoot, PO
- ▶ Settling time, T_s
- ▶ Steady-state error, e_{ss}



Peak Time, T_p

The **peak time** is the time at which the response first reaches its maximum value.
For the underdamped step response,

$$\dot{y}(t) = \frac{\omega_n}{\beta} e^{-\zeta\omega_n t} \sin(\beta\omega_n t).$$

At the peak,

$$\dot{y}(T_p) = 0.$$

Thus,

$$\sin(\beta\omega_n T_p) = 0.$$

For the first positive peak,

$$\beta\omega_n T_p = \pi.$$

Peak Time

$$T_p = \frac{\pi}{\beta\omega_n} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

Maximum Overshoot

At

$$T_p = \frac{\pi}{\beta\omega_n},$$

the maximum value of the response is

$$M_p = y(T_p).$$

Substituting T_p into the step-response equation gives

$$M_p = 1 + e^{-\zeta\pi/\beta}.$$

Therefore, the amount by which the response exceeds its final value is

$$M_p - 1 = e^{-\zeta\pi/\beta}.$$

Since $\beta = \sqrt{1 - \zeta^2}$,

$$M_p - 1 = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$$

Percent Overshoot, PO

The **percent overshoot** expresses the maximum overshoot as a percentage of the final value:

$$PO = (M_p - 1) \times 100.$$

For a standard underdamped second-order system,

$$PO = 100e^{-\zeta\pi/\beta}.$$

Since $\beta = \sqrt{1 - \zeta^2}$, we obtain

Percent Overshoot

$$PO = 100 \exp \left(-\frac{\zeta\pi}{\sqrt{1 - \zeta^2}} \right)$$

- ▶ Increasing ζ produces **less overshoot**.
- ▶ As $\zeta \rightarrow 1$, the oscillatory overshoot disappears.

Settling Time, T_s

The **settling time** is the time required for the response to enter and remain within a specified tolerance band around its final value.

For a 2% criterion, use the decay envelope

$$e^{-\zeta\omega_n T_s} = 0.02.$$

Taking logarithms,

$$-\zeta\omega_n T_s = \ln(0.02).$$

Thus,

$$T_s = \frac{-\ln(0.02)}{\zeta\omega_n}.$$

Since

$$-\ln(0.02) \approx 3.912,$$

$$T_s \approx \frac{3.912}{\zeta\omega_n}.$$

The common approximation is

2% Settling Time

$$T_s \approx \frac{4}{\zeta\omega_n}$$

Rise Time, T_r

The **rise time** measures how rapidly the output approaches its final value.

- ▶ A common definition is the time required for the response to rise from 10% to 90% of its final value.
- ▶ Let

$$y(t_{10}) = 0.1, \quad y(t_{90}) = 0.9.$$

- ▶ Then,

$$T_r = t_{90} - t_{10}$$

- ▶ For the underdamped second-order response,

$$y(t) = 1 - e^{-\zeta\omega_n t} \left[\cos(\beta\omega_n t) + \frac{\zeta}{\beta} \sin(\beta\omega_n t) \right],$$

the values t_{10} and t_{90} are obtained from the corresponding crossing times.

Steady-State Error

The **steady-state error** measures the difference between the desired final value and the actual final output:

$$e_{ss} = \text{Desired steady-state value} - \text{Actual steady-state value.}$$

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Using the final value theorem,

Hence,

$$y(\infty) = \lim_{s \rightarrow 0} sY(s).$$

$$e_{ss} = 1 - \lim_{s \rightarrow 0} G(s)$$

For a unit-step input, $U(s) = \frac{1}{s}$, so

For

$$Y(s) = G(s)\frac{1}{s}.$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

Therefore,

$$y(\infty) = \lim_{s \rightarrow 0} G(s).$$

$$T(0) = 1 \Rightarrow e_{ss} = 0.$$

Second-Order Performance Specifications

For a standard underdamped second-order system,

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad 0 < \zeta < 1.$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$T_s \approx \frac{4}{\zeta\omega_n} \quad (2\%)$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$T_r = t_{90} - t_{10}$$

$$PO = 100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$e_{ss} = 1 - \lim_{s \rightarrow 0} G(s)$$

Thank You

Questions and Discussion